

# Aggregation Semantics for Prioritization in Cooperative Automated Decision Making

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## Abstract

To be rewritten.

**Keywords:** cooperative automated decision making; aggregation operators; multi-criteria decision making; context-aware decision support; prioritization

## 1. Introduction

## 2. Background and Related Work

This section provides the conceptual foundations required for the remainder of the paper. We first review the literature on context-aware decision support systems and aggregation operators, highlighting the gap between the representation of contextual knowledge and the analysis of the aggregation mechanisms used to combine contextual and predictive information. We then introduce the Cooperative Automated Decision-Making System (CADEMAS) framework and its machine-learning instantiation (CADEMAS-ML), which provide the architectural setting in which the aggregation strategies studied in this work are analyzed.

### 2.1. Related Work

Research on cooperative and context-aware decision support has established multiple mechanisms for incorporating human knowledge into data-driven systems. These include symbolic rules, ontologies, clinical pathways, and human-centred interaction principles. For example, Teng and Tan [25] proposed cognitive agents that combine propositional rules with reinforcement learning to support context-aware decision making. In the clinical domain, Yao and Kumar [26] employed ontologies to identify context-appropriate rules and instantiate clinical-pathway fragments for individual patients. More recently, Cronholm and Göbel [27] argued that effective decision support requires combining human and artificial intelligence through appropriate interaction and governance mechanisms rather than relying solely on predictive performance. Complementing these studies, the survey of Merkert et al. [30] found that the usefulness of machine learning in decision support depends strongly on the decision task, process stage, and organizational context, while human and organizational evaluation criteria remain comparatively underexplored.

Collectively, these works demonstrate the importance of contextual knowledge in decision support systems. However, their primary focus is typically on knowledge representation, architecture design, rule selection, or human–AI interaction. Comparatively less attention has been paid to the numerical integration step that combines predictive evidence with contextual assessments once both sources of information are available. As a result,

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simple weighted averages are often adopted as a convenient integration mechanism, even though their compensatory semantics may allow strong predictive evidence to offset weak contextual support. While contextual rules can impose constraints upstream, this does not necessarily guarantee that such constraints are preserved after score fusion.

The aggregation literature provides the mathematical foundations for analyzing this issue. Among the most influential contributions, Yager's ordered weighted averaging (OWA) operator [7] introduced a family of aggregation functions capable of representing different degrees of conjunctive and disjunctive behaviour through ordered weights. Yager [28] subsequently extended this framework by combining OWA operators with t-norms and relating them to Choquet and Sugeno integrals. More broadly, aggregation theory has shown that the choice of operator determines the degree of compensation allowed among criteria and therefore influences the resulting decision semantics. This importance is reflected in the review by Mardani et al. [13], which identified OWA-based methods as one of the most widely used aggregation families in fuzzy MCDM applications. Similarly, Csiszár [31] characterized OWA as a flexible parameterized framework spanning numerous well-known operators while highlighting the challenge of selecting appropriate aggregation behaviour.

A related line of research has investigated aggregation in collective decision-making settings. Moral et al. [29] showed that both the aggregation operator and the associated distance measure affect the speed and quality of consensus formation. Trillo et al. [32] further explored customizable aggregation mechanisms that allow experts to express different attitudes toward criterion importance and consensus requirements. These studies provide valuable insights into compensation, consensus, and preference aggregation. Nevertheless, their focus remains on combining preferences or judgments from multiple decision makers rather than integrating predictive outputs from machine-learning models with contextual signals that encode organizational priorities, regulations, or policy constraints.

Consequently, a gap remains between context-aware decision-support frameworks and aggregation theory. The former provides mechanisms for representing and managing contextual knowledge but rarely examines the behavioral implications of alternative aggregation semantics. The latter offers a rich collection of aggregation operators and theoretical properties but has seldom been studied from the perspective of policy-compliant score fusion in cooperative automated decision-making architectures. In particular, relatively little attention has been paid to questions such as whether contextual assessments can effectively act as veto mechanisms, how aggregation semantics influence robustness to predictive overconfidence, or how prioritization rankings respond to uncertainty in predictive and contextual information. These questions motivate the analysis developed in this work. Table 1 summarizes representative contributions along these dimensions and highlights the gap addressed here.

Unlike traditional fuzzy MCDM studies, the objective of this work is not to propose a new aggregation operator, derive additional algebraic properties, or solve a specific multicriteria decision problem through a particular aggregation scheme. Instead, aggregation operators are treated as alternative policy semantics within a cooperative automated decision-making architecture. Likewise, unlike many context-aware DSS frameworks, the focus is not on representing contextual knowledge itself but on understanding how different integration mechanisms affect the interaction between predictive evidence and contextual constraints. Consequently, the contribution of this work lies at the architectural and methodological level, namely, the systematic analysis of aggregation as a governance mechanism within CADEMAs-ML.

**Table 1.** Chronological comparison of representative related work. “Partially” indicates that a contribution can encode rules, conjunctive criteria, or minimum thresholds, but does not establish veto preservation for predictive–contextual score fusion.

| Reference                      | Application / Framework                | Aggregation method                                                                    | Strict contextual veto | Operator robustness |
|--------------------------------|----------------------------------------|---------------------------------------------------------------------------------------|------------------------|---------------------|
| Yager [28] (2005)              | Multicriteria decision making          | TOWA (t-norm–OWA); Choquet and Sugeno extensions                                      | Partially              | Partially           |
| Teng and Tan [25] (2008)       | Context-aware DSS via cognitive agents | Symbolic rules with reinforcement learning; scalar fusion not specified               | Partially              | No                  |
| Yao and Kumar [26] (2012)      | Ontology-based clinical DSS            | Context-driven rule / pathway selection; scalar fusion not specified                  | Partially              | No                  |
| Merkert et al. [30] (2015)     | Survey of ML in DSS                    | Not a specific aggregation operator                                                   | No                     | No                  |
| Moral et al. [29] (2017)       | Group decision making and consensus    | Aggregation operators combined with distance functions                                | No                     | Partially           |
| Mardani et al. [13] (2018)     | Review of fuzzy MCDM                   | Reviews OWA and related fuzzy aggregation families                                    | No                     | No                  |
| Csiszár [31] (2021)            | OWA theory and applications            | Parameterized OWA family; weight selection                                            | Partially              | Partially           |
| Cronholm and Göbel [27] (2022) | Human-centred AI decision support      | Human–AI design principles; scalar fusion not specified                               | No                     | No                  |
| Trillo et al. [32] (2023)      | Consensus-based multicriteria GDM      | Expert-customized operator; per-criterion minimum consensus                           | Partially              | No                  |
| Present work                   | Cooperative ADM (CADEMAS)              | Comparative linear, geometric, and minimum fusion of predictive and contextual scores | Yes                    | Yes                 |

2.2. CADEMAs and CADEMAs-ML

The literature reviewed in the previous section shows that many context-aware decision-support systems focus on representing contextual knowledge through rules, ontologies, policies, or human expertise. However, comparatively less attention has been paid to the integration mechanism that combines contextual assessments with predictive evidence. The CADEMAs framework introduced by Novoa-Hernández et al. [4] provides a suitable setting for studying this issue because it explicitly separates evidence generation from contextual evaluation and decision integration.

CADEMAS is a general framework for designing cooperative automated decision-making systems in which multiple heterogeneous Automated Decision-Making (ADM) units collaborate to produce a unified recommendation, score, or label. The framework is intentionally agnostic with respect to the internal implementation of these units. An ADM may correspond to a machine-learning model, a rule-based system, an optimization algorithm, an expert system, or any other computational mechanism capable of generating decision evidence. Likewise, contextual knowledge may be represented through expert rules, organizational policies, regulatory requirements, or other knowledge-representation formalisms.

Rather than prescribing how individual decisions are produced, CADEMAs defines an architectural process through which heterogeneous decision signals are coordinated and

subsequently evaluated against contextual considerations before a final recommendation is generated.

Formally, a CADEMAs instance is represented by

$$\mathcal{S} = \langle I, U, O, K, T \rangle, \quad (1)$$

where  $I$  denotes the Input Manager,  $U$  the set of ADM units,  $O$  the Decision Integration Layer,  $K$  the Contextual Modulator, and  $T$  the universe of admissible data representations.

The resulting architecture follows a layered decision process. First, the ADM units independently generate decision evidence from the available information. Different units may operate on different subsets of the input data and employ different reasoning mechanisms. In parallel, the Contextual Modulator evaluates the same decision case from the perspective of organizational priorities, domain knowledge, regulatory requirements, or other contextual considerations, producing a degree of contextual satisfaction. Finally, the Decision Integration Layer combines both sources of information into a unified decision output.

This explicit separation between predictive assessment, contextual evaluation, and decision integration is one of the defining characteristics of CADEMAs. It allows the same architecture to be instantiated across different domains while making the integration mechanism itself an independent design component.

A particularly relevant specialization is CADEMAs-ML, in which the ADM units correspond to machine-learning models. In this case, the predictive subsystem resembles a cooperative ensemble in which multiple models contribute to a unified predictive assessment. Unlike conventional ensemble methods, however, CADEMAs-ML separates predictive cooperation from contextual evaluation. Predictive outputs are first consolidated into a common predictive score and only afterwards combined with contextual knowledge that encodes organizational or institutional priorities. Following previous CADEMAs-ML implementations [6,20,21], contextual knowledge is represented through an interpretable fuzzy inference system.

To focus on the role of aggregation independently of any specific application domain, we abstract the predictive subsystem by a normalized predictive score

$$R_i \in [0, 1], \quad (2)$$

which summarizes the evidence generated by the cooperating ADM units for decision case  $i$ . Depending on the application,  $R_i$  may represent a risk estimate, a confidence score, a probability of occurrence, or any other normalized predictive assessment.

Independently, the contextual subsystem produces a normalized context score

$$Q_i \in [0, 1], \quad (3)$$

which measures the extent to which decision case  $i$  satisfies the relevant contextual requirements, policies, or priorities.

The predictive and contextual assessments are then combined through an aggregation operator  $A$

$$P_i = A(R_i, Q_i), \quad (4)$$

yielding the final prioritization score. The resulting scores are used to rank candidate cases and determine the corresponding Top- $K$  alternatives.

Existing CADEMAs-ML applications employ a linear convex aggregation rule to combine predictive and contextual information. However, the CADEMAs framework itself does not prescribe any specific aggregation semantics [4]. Consequently, the aggregation

operator can be regarded as an independent policy-design component that determines how predictive evidence and contextual considerations interact. This property makes CADEMAs-ML an appropriate testbed for investigating how alternative aggregation semantics influence prioritization behavior, policy compliance, robustness, and ranking stability while preserving the same predictive and contextual subsystems.

### 3. Theoretical Framework and Aggregation Properties

#### 3.1. Preliminaries and Assumptions

Let  $R \in [0, 1]$  represent the predictive score and  $Q \in [0, 1]$  represent the contextual score assigned to a given decision alternative. While our analysis assumes these scores are normalized to the unit interval, this assumption does not restrict the universality of our findings. Any bounded continuous scoring domain  $[a, b]$  can be isomorphically mapped to  $[0, 1]$  through standard min-max scaling. Consequently, the theoretical properties and operator behaviors discussed herein hold universally for any bounded decision space.

In this framework, the integration of predictive and contextual dimensions is governed by an *aggregation function*. Formally, a bivariate aggregation function is a mapping  $A : [0, 1]^2 \rightarrow [0, 1]$  that satisfies the following fundamental axioms:

- *Boundary conditions:*  $A(0, 0) = 0$  and  $A(1, 1) = 1$ .
- *Monotonicity:*  $A(x_1, y_1) \leq A(x_2, y_2)$  whenever  $x_1 \leq x_2$  and  $y_1 \leq y_2$ .

Depending on the chosen mechanism, additional algebraic properties may be desirable, such as commutativity ( $A(x, y) = A(y, x)$ ) and idempotency ( $A(x, x) = x$ ). It is crucial to distinguish between general aggregation functions and specific mathematical sub-families. For example, while triangular norms (t-norms) such as the standard minimum operator are a specific class of aggregation functions that strictly require commutativity and associativity, not all operators analyzed in cooperative frameworks belong to this class.

Building upon these standard axioms, we establish the following baseline lemma, which governs the structural behavior of any valid aggregation process within our system.

**Lemma 1. Lemma 1 (Monotonicity and Boundary Preservation).** *For any candidate evaluated under an aggregation function  $A$  satisfying the standard axioms, an isolated improvement in either the predictive score  $R$  or the contextual score  $Q$  guarantees that the final integrated score will not decrease. Furthermore, the evaluation is strictly bounded such that the presence of a perfect partial score cannot force the aggregation beyond the strict limits of the codomain, meaning  $A(R, 0) \geq 0$  and  $A(R, 1) \leq 1$  for any  $R \in [0, 1]$ .*

#### 3.2. Choice of Operators and Archetypes

#### 3.3. Veto Preservation and Compensation Limits

#### 3.4. Sensitivity and Robustness to Contextual Imprecision

### 4. Experimental Evaluation: Monte Carlo Simulations

#### 4.1. Experimental Protocol and Metrics

#### 4.2. Compensation Effects under Intermediate Contexts

#### 4.3. Robustness to Contextual Uncertainty

### 5. Real-World Case Study: Employee Attrition Prioritization

#### 5.1. Dataset Setup and Contextual Vetoes

#### 5.2. Prioritization Results and Top-K Stability

### 6. Discussion and Conclusions

**Author Contributions:** To be completed.



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